

Each criterion was evaluated against every other using a pairwise comparison method. Values were assigned in 0.25 increments between 0 and 1, where:

- A value of 0.5 indicates equal importance.
- Values of 0.25 or 0.75 indicate one criterion is moderately more important than the other.
- Values of 0 or 1 reflect extreme preference toward one criterion, rendering the other negligible.

The comparisons are symmetric such that the sum of each pair equals 1. Total weights were summed and normalized to create priority values for each criterion.

| Criteria                  | Mass | Power Demand | Data Rate | Pointing & Ops | Reliability/TRL | Weight | Normalized |
|---------------------------|------|--------------|-----------|----------------|-----------------|--------|------------|
| Mass                      |      | 0.25         | 0.25      | 0.5            | 0.25            | 1.25   | 0.13       |
| Power Demand              | 0.75 |              | 0.5       | 0.75           | 0.5             | 2.5    | 0.25       |
| Data Rate Capability      | 0.75 | 0.5          |           | 0.75           | 0.5             | 2.5    | 0.25       |
| Pointing & Ops Complexity | 0.5  | 0.25         | 0.25      |                | 0.25            | 1.25   | 0.13       |
| Reliability/TRL           | 0.75 | 0.5          | 0.5       | 0.75           |                 | 2.5    | 0.25       |
| Total                     |      |              |           |                |                 | 10     |            |

|   | Mass    | Power Demand | Data Rate         | Pointing & Ops           | Reliability/TRL          |
|---|---------|--------------|-------------------|--------------------------|--------------------------|
| 5 | < 3 kg  | < 25 W       | > 2 Mbps          | Hemispherical; no        | TRL 8-9; flight-proven   |
| 4 | 3-5 kg  | 25-50 W      | 512 kbps - 2 Mbps | Fixed; ±30° usable beam  | TRL 7; minor             |
| 3 | 5-8 kg  | 50-75 W      | 128-512 kbps      | Body-pointed or single-  | TRL 5-6; unqualified for |
| 2 | 8-12 kg | 75-100 W     | 32-128 kbps       | Coarse two-axis pointing | TRL 4-5; prototype only  |
| 1 | > 12 kg | > 100 W      | < 32 kbps         | Closed-loop tracked      | TRL < 3                  |

| Analysis of Rover-to-Relay Proximity Link Configuration |        |                                           |                |                              |                |                                           |                |
|---------------------------------------------------------|--------|-------------------------------------------|----------------|------------------------------|----------------|-------------------------------------------|----------------|
| Decision Criteria                                       |        | UHF quadrifilar helix (401.6 / 437.1 MHz) |                | S-band patch array (2.2 GHz) |                | X-band steerable HGA cross-link (8.4 GHz) |                |
| Criteria                                                | Weight | Score                                     | Weighted Score | Score                        | Weighted Score | Score                                     | Weighted Score |
| Mass                                                    | 0.13   | 5                                         | 0.65           | 4                            | 0.52           | 2                                         | 0.26           |
| Power Demand                                            | 0.25   | 4                                         | 1.00           | 3                            | 0.75           | 2                                         | 0.50           |
| Data Rate Capability                                    | 0.25   | 4                                         | 1.00           | 5                            | 1.25           | 5                                         | 1.25           |
| Pointing & Ops Complexity                               | 0.13   | 5                                         | 0.65           | 4                            | 0.52           | 1                                         | 0.13           |
| Reliability/TRL                                         | 0.25   | 5                                         | 1.25           | 3                            | 0.75           | 2                                         | 0.50           |
| Total Score                                             |        | 4.55                                      |                | 3.79                         |                | 2.64                                      |                |

The UHF quadrifilar helix scores 4.55 against 3.79 for the S-band patch array and 2.64 for the X-band gimbaled cross-link. The X-band option has the raw throughput but fails the other two criteria: a two-axis gimbal is a moving mechanism that has to survive 90 K and 100 krad, and it needs closed-loop tracking of a relay the rover cannot see coming. S-band buys data rate the mission does not need, since the link budget shows the UHF return already closes at 128 kbps with 18.3 dB of margin and is bounded by the transceiver's 2048 kbps ceiling rather than by RF. The helix is also already in the mass budget at 0.86 kg and carries TRL 9 through Electra and Electra-Lite on exactly this kind of proximity link. Selected: UHF quadrifilar helix.